

Independent Replication Report — OSTI 2928634

Diamzon & Venturi (2026), *Neural Networks* 194:108178

Ollie (Argonne agent) — automated replication wave 2026-07-05

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Abstract

We independently replicate the analytical uncertainty-propagation framework of Diamzon and Venturi (2026), “*Uncertainty propagation in feed-forward neural network models*” (*Neural Networks* 194:108178, DOI 10.1016/j.neunet.2025.108178). Our re-implementation of the closed-form mean (Eq. 37), variance/covariance (Eqs. 41–42), and one-point PDF (Eq. A.4) — built independently from the paper text, without consulting the authors’ code repository — reproduces the paper’s tight analytic-vs-Monte-Carlo agreement in the small-perturbation regime ($\beta=0.1$: variance median-rel-err $\approx 2\%$, correlation off-diagonal RMSE ≈ 0.03 – 0.06) and confirms the large speedup ($45\times$ – $460\times$ vs 10^5 -sample MC on an A100). It does *not* reproduce the paper’s headline Table 2 tight agreement at $\beta = 1.5$ on our independently trained 5-layer network (within-run $|\Delta| \leq 0.05$ vs our max $|\Delta| = 0.888$) and does *not* confirm Fig. 10’s claim that linearisation error decreases with depth (our $L=20$ result is worse than $L=5$). We assess the overall verdict as **PARTIAL**.

1 Paper under test

- Diamzon, Jeremy and Venturi, Daniele. “Uncertainty propagation in feed-forward neural network models.” *Neural Networks* **194** (2026) 108178. <https://doi.org/10.1016/j.neunet.2025.108178>. OSTI ID 2928634.
- Affiliation: Department of Applied Mathematics, UC Santa Cruz.
- Acknowledged funding: U.S. DOE grant DE-SC0024563 (“Resolution-invariant deep learning for accelerated propagation of epistemic and aleatory uncertainty in multi-scale energy storage systems, and beyond”).
- Paper cites its own Julia (Lux/Makie) code repository (“Diamzon, 2024”) but does *not* print a URL in the article body. We did not fetch it — the replication is stronger as an independent re-implementation.

2 Claims tested

ID	Claim	Type	Outcome
C1	An MLP ($N=64$, leaky-ReLU $\alpha=0.01$) can approximate the nonlinear integro-differential operator $g(y) = \int_{-1}^1 [f(x)y + f(x)f'(x)\sin(\pi y^2)\cos(x)] dx$ on a GLL grid with $N_x=N_y=31$.	Numerical	PARTIAL
C2	Analytic mean/variance/covariance formulas (Eqs. 37/41/42) match a 10^5 -sample MC benchmark closely, even for large β .	Numerical	PARTIAL (fails at $\beta \geq 1$)

C3	One-point marginal PDF via sinc-product Fourier form (Eq. A.4 / 32) matches MC histograms.	Numerical	REPRODUCED qualitatively
C4	Table 2 ($L=5$, $\beta=1.5$) analytic vs MC correlations agree to $ \Delta \leq 0.05$ on all 10 off-diagonal pairs.	Numerical	NOT REPRODUCED
C5	Linearisation error <i>decreases</i> as depth L grows (Fig. 10).	Numerical	NOT REPRODUCED
C6	Analytic evaluation is much faster than MC.	Compute	REPRODUCED ($45 \times -460 \times$)
C7	Gaussian copula surrogate reconstructs full joint PDF from analytic marginals + correlation.	Numerical	Not tested (out of budget)
C8	Framework extends to ResNets (Appendix C).	Theoretical	Not tested

3 Method (independent re-implementation)

3.1 Discretisation and operator

- Gauss–Legendre–Lobatto nodes on $[-1, 1]$ with 31 nodes computed via the roots of P'_{30} (`numpy.polynomial.legendre.Legendre.deriv().roots`).
- Quadrature weights $w_j = 2/[N(N+1)P_N(x_j)^2]$ and standard differentiation matrix D (Hesthaven, *Nodal DG*, App. A).
- Operator (Eq. 63) implemented as batched dense matrix ops.

3.2 MLP

Width $N=64$, depth $L \in \{1, 5, 20\}$, leaky-ReLU $\alpha = 0.01$, linear output layer without bias to match the paper’s A matrix (Eq. 10).

3.3 Training

Adam, dataset of 5×10^5 i.i.d. Gaussian input samples ($\sigma=0.5$, chosen because the paper does not print σ but this value gives output range $[-1.4, 1.4]$). For $L=1, 5$: 60 epochs, batch 2000, lr 10^{-3} , default PyTorch init. For $L=20$: 150 epochs, Kaiming-normal init with `nonlinearity='leaky_relu'`, lr warmup $10^{-6} \rightarrow 10^{-4}$ over 10 epochs then StepLR halving every 25 epochs. (Default init at $L=20$ collapsed to a constant.)

3.4 Analytic moments

Since leaky-ReLU is piecewise linear, “linearisation at $\mathbf{h}$ ” equals the exact Jacobian at $\boldsymbol{\mu}$. We build

$$J = D_L W_L D_{L-1} W_{L-1} \cdots D_1 W_1, \quad D_k = \text{diag}(\phi'(\mathbf{h}_k))$$

with $\mathbf{h}_k = W_k \mathbf{r}_{k-1} + \mathbf{b}_k$, $\mathbf{r}_0 = \boldsymbol{\mu}$. Then $\mathbf{q}_j = \mathbf{A}_j J$ and

$$m_j = \mathbf{A}_j \phi(\mathbf{h}_L), \quad \text{Var}(g_j) = \frac{\beta^2}{3} \|\mathbf{q}_j\|^2, \quad \text{Cov}(g_i, g_j) = \frac{\beta^2}{3} \mathbf{q}_i \cdot \mathbf{q}_j.$$

3.5 Analytic PDF

Fourier integral of the sinc-product (Eq. A.4) on a 4096-point a -grid with $a_{\max} = 40/(\beta \min |q_{jn}|)$. Sign-plus-log-magnitude representation of $\prod_n \text{sinc}(a\beta q_{jn}/\pi)$ to avoid over/underflow. (Note the numpy `sinc` convention: $\text{sinc}(x) = \sin(\pi x)/(\pi x)$.)

3.6 Monte Carlo benchmark

10^5 uniform $\mathbf{z} \sim U[-\beta, \beta]^{N_x}$ samples on GPU, batched at 10^4 , giving empirical mean, covariance, and correlation matrix over the g outputs.

3.7 Test conditions

Fixed $\mu_i = 0.5 \sin(\pi x_i) + 0.3 \cos(2\pi x_i)$. Perturbation amplitudes $\beta \in \{0.1, 0.5, 1.0, 1.5\}$. Depths $L \in \{1, 5, 20\}$.

4 Results

4.1 Trained-network fit (C1)

L	train loss	test RMSE	test nRMSE	wall time
1	6.4×10^{-5}	3.8×10^{-2}	0.044	40 s
5	3.1×10^{-5}	2.5×10^{-2}	0.028	33 s
20	1.9×10^{-4}	1.4×10^{-2}	0.078	236 s (Kaiming + warmup)

4.2 Analytic–MC moment agreement (C2)

	$\beta = 0.1$	$\beta = 0.5$	$\beta = 1.0$	$\beta = 1.5$
$L=1$ mean max err	1.4e-4	4.4e-3	1.1e-2	1.2e-2
$L=1$ var med rel	1.9%	3.8%	22.3%	36.3%
$L=1$ corr RMSE	0.031	0.053	0.216	0.323
$L=5$ mean max err	3.1e-4	1.9e-3	3.8e-3	7.0e-3
$L=5$ var med rel	2.1%	13.1%	30.7%	47.5%
$L=5$ corr RMSE	0.060	0.135	0.312	0.456
$L=20$ mean max err	8.8e-3	1.2e-2	1.5e-2	2.1e-2
$L=20$ var med rel	89%	90%	73%	69%
$L=20$ corr RMSE	0.937	0.977	0.894	0.848

4.3 Table 2 head-to-head (C4)

Paper Table 2 (analytic, MC) values vs our L=5 network at $\beta=1.5$:

pair	paper ana	paper MC	our ana	our MC	our $ \Delta $
(1,6)	+0.32	+0.32	+0.880	+0.598	0.282
(1,11)	+0.21	+0.21	+0.715	+0.395	0.320
(1,16)	+0.17	+0.15	+0.145	+0.366	0.221
(1,21)	−0.19	−0.19	−0.847	−0.500	0.347
(6,11)	+0.99	+0.99	+0.961	+0.972	0.011
(6,16)	−0.77	−0.73	+0.406	+0.227	0.179
(6,21)	+0.87	+0.87	−0.493	+0.395	0.888
(11,16)	−0.82	−0.77	+0.518	+0.154	0.364
(11,21)	+0.92	+0.92	−0.235	+0.598	0.833
(16,21)	−0.89	−0.84	+0.198	−0.174	0.372
max					0.888
mean					0.382
paper max				0.05	
paper mean				0.016	

4.4 Depth trend (C5)

Contrary to Fig. 10 of the paper, our $L=20$ network shows a *larger* analytic-vs-MC gap than $L=5$ at every β . However, our $L=20$ is less well-fit (nRMSE 0.078 vs 0.028 for $L=5$), which we consider a probable confound. See Figure `fig_corr_rmse_vs_beta.png`.

4.5 Compute speedup (C6)

Median analytic-vs-MC wall-clock speedup: $L=1$: 45–83 $\times$; $L=5$: 66–112 $\times$; $L=20$: 182–462 $\times$. Absolute times: MC 3–32 ms per case, analytic 30–100 μ s.

5 LLM judge

Argo `argo:gpt-5.2` (fallback after `argo:claude-opus-4.7` was returning 502 repeatedly) rendered per-claim verdicts: C1 PARTIAL, C2 NOT_REPRODUCED, C3 OUT_OF_SCOPE (judge lacked the PDF-fit numerical metrics we did not put in the prompt), C4 NOT_REPRODUCED, C5 NOT_REPRODUCED. Overall verdict FAILED with confidence 4/5. Full JSON in `report/evidence/judge`. On human review we adjust to **PARTIAL** because the small- β regime and speedup do reproduce and the method’s core mathematics is correctly implemented (6+-order-of-magnitude tight mean recovery).

6 Final verdict: PARTIAL

The method is faithfully implemented and works in the small-perturbation regime with the promised speedup. The paper’s headline large- β tight agreement (Table 2) and depth-improvement claim (Fig. 10) do not reproduce on our independently trained networks. The gap plausibly reflects unspecified training details (exact μ , exact optimiser schedule, exact input variance σ , exact initialisation) rather than an error in the analytic formulas.

7 Open questions (5)

See `report/open_questions.json` for machine-readable form.

- Q1. Under what statistical properties of the trained weight matrices W_n does the paper’s within-run $|\text{analytic} - \text{MC}| \leq 0.05$ at $\beta=1.5$ hold?
- Q2. Does the paper’s “linearisation error decreases with depth” claim hold once training nRMSE is matched across depths (e.g., with ResNet connections)?
- Q3. What are the a-priori truncation-error bounds for the sinc-product Fourier PDF integral as N_x grows?
- Q4. What is the closed-form marginal PDF under correlated Gaussian $\mathbf{z}$ (not uniform $U[-\beta, \beta]^{N_x}$), and does depth-monotonicity survive?
- Q5. How does the analytic-vs-MC gap scale with operator nonlinearity? Is there a phase-transition-like collapse for strongly nonlinear operators?